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Ex No – 6A	First Come And First Serve

Program :

```
#include <stdio.h>
int main() {
  int n, i;
  float totalWaitingTime = 0, totalTurnaroundTime = 0;
  printf("Enter the number of processes: ");
  scanf("%d", &n);
  int burstTime[n], waitingTime[n], turnaroundTime[n];
  printf("Enter the burst time of the processes (separated by space): \n");
  for(i = 0; i < n; i++) {
    scanf("%d", &burstTime[i]);
  }
  waitingTime[0] = 0;
  turnaroundTime[0] = burstTime[0];
  for(i = 1; i < n; i++) {
    waitingTime[i] = waitingTime[i-1] + burstTime[i-1];
    turnaroundTime[i] = waitingTime[i] + burstTime[i];
  }
  printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
  for(i = 0; i < n; i++) {
    printf("%d\t\t%d\t\t%d\t\t%d\n", i, burstTime[i], waitingTime[i], turnaroundTime[i]);
    totalWaitingTime += waitingTime[i];
    totalTurnaroundTime += turnaroundTime[i];
  }
  printf("\nAverage waiting time is: %.2f\n", totalWaitingTime / n);
```

```c
    printf("Average turnaround time is: %.2f\n", totalTurnaroundTime / n);

    return 0;

}
```

Output :

```
Enter the number of processes: 3
Enter the burst time of the processes (separated by space):
24 3 3

Process Burst Time      Waiting Time    Turnaround Time
0             24              0               24
1             3               24              27
2             3               27              30

Average waiting time is: 17.00
Average turnaround time is: 27.00
[cse32@localhost ~]$
```

Ex No – 6B	Shortest Job First

Program :

```c
#include <stdio.h>

struct Process {

    int id, burst_time, waiting_time, turnaround_time;

};

int main() {

    int n;

    printf("Enter number of processes:\n");

    scanf("%d", &n);

    struct Process proc[n];

    printf("Enter burst times:\n");

    for (int i = 0; i < n; i++) {

        proc[i].id = i + 1;

        scanf("%d", &proc[i].burst_time);

    }

    for (int i = 0; i < n - 1; i++) {

        for (int j = i + 1; j < n; j++) {

            if (proc[i].burst_time > proc[j].burst_time) {
```

```
            struct Process temp = proc[i];
            proc[i] = proc[j];
            proc[j] = temp;
          }
        }
    }
    proc[0].waiting_time = 0;
    for (int i = 1; i < n; i++) {
        proc[i].waiting_time = proc[i - 1].waiting_time + proc[i - 1].burst_time;
    }
    for (int i = 0; i < n; i++) {
        proc[i].turnaround_time = proc[i].waiting_time + proc[i].burst_time;
    }
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    int total_waiting_time = 0, total_turnaround_time = 0;
    for (int i = 0; i < n; i++) {
        printf("%d\t%d\t\t%d\t\t%d\n", proc[i].id, proc[i].burst_time, proc[i].waiting_time,
proc[i].turnaround_time);
        total_waiting_time += proc[i].waiting_time;
        total_turnaround_time += proc[i].turnaround_time;
    }
    printf("\nAverage Waiting Time: %.2f\n", (float)total_waiting_time / n);
    printf("Average Turnaround Time: %.2f\n", (float)total_turnaround_time / n);
    return 0;
}
```

Output :

```
Enter number of processes:
4
Enter burst times:
8 4 9 5

Process Burst Time      Waiting Time    Turnaround Time
2       4               0               4
4       5               4               9
1       8               9               17
3       9               17              26

Average Waiting Time: 7.50
Average Turnaround Time: 14.00
[cse32@localhost ~]$
```

Ex No – 6C	Priority Scheduling

Program :

```
#include <stdio.h>

struct Process {
    char name[5];
    int burst_time;
    int priority;
    int waiting_time;
    int turnaround_time;
};

void sortProcesses(struct Process processes[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (processes[j].priority > processes[j + 1].priority) {
                struct Process temp = processes[j];
                processes[j] = processes[j + 1];
                processes[j + 1] = temp;
            }
        }
    }
}

int main() {
    int n;
```

```
    printf("Enter the number of processes: ");
    scanf("%d", &n);
    printf("Enter Burst Time and Priority");
    struct Process processes[n];
    for (int i = 0; i < n; i++) {
        printf("\nP[%d]\n", i + 1);
        printf("Burst time: ");
        scanf("%d", &processes[i].burst_time);
        printf("Priority: ");
        scanf("%d", &processes[i].priority);
        sprintf(processes[i].name, "P[%d]", i + 1);
    }
    sortProcesses(processes, n);
    int total_waiting_time = 0, total_turnaround_time = 0;
    for (int i = 0; i < n; i++) {
        if (i == 0) {
            processes[i].waiting_time = 0;
} else {
            processes[i].waiting_time = processes[i - 1].waiting_time + processes[i - 1].burst_time;
        }
        processes[i].turnaround_time = processes[i].waiting_time + processes[i].burst_time;
        total_waiting_time += processes[i].waiting_time;
        total_turnaround_time += processes[i].turnaround_time;
    }
    printf("\nProcess\tBurst Time\tPriority\tWaiting Time\tTurnaround Time\n");
    for (int i = 0; i < n; i++) {
        printf("%s\t%d\t\t%d\t\t%d\t\t%d\n",
               processes[i].name,
               processes[i].burst_time,
               processes[i].priority,
               processes[i].waiting_time,
```

```
            processes[i].turnaround_time);
    }
    printf("\nAverage Waiting Time: %d", total_waiting_time / n);
    printf("\nAverage Turnaround Time: %d\n", total_turnaround_time / n);
    return 0;
}
```

Output :

```
Enter the number of processes: 4
Enter Burst Time and Priority
P[1]
Burst time: 6
Priority: 3

P[2]
Burst time: 2
Priority: 2

P[3]
Burst time: 14
Priority: 1

P[4]
Burst time: 6
Priority: 4

Process Burst Time      Priority        Waiting Time    Turnaround Time
P[3]    14              1               0               14
P[2]    2               2               14              16
P[1]    6               3               16              22
P[4]    6               4               22              28

Average Waiting Time: 13
Average Turnaround Time: 20
[cse32@localhost ~]$
```

Ex No – 6D	Round Robin Scheduling

Program :

```
#include <stdio.h>
#include <stdbool.h>
```

```
struct Process {
    int id;
    int arrival_time;
    int burst_time;
    int remaining_time;
    int waiting_time;
    int turnaround_time;
};
void calculate_times(struct Process processes[], int n, int quantum) {
    int time = 0;
    bool done;
    do {
        done = true;
        for (int i = 0; i < n; i++) {
            if (processes[i].remaining_time > 0) {
                done = false;
                if (processes[i].remaining_time > quantum) {
                    time += quantum;
                    processes[i].remaining_time -= quantum;
                } else {
                    time += processes[i].remaining_time;
                    processes[i].waiting_time = time - processes[i].burst_time - processes[i].arrival_time;
                    processes[i].remaining_time = 0;
                }
            }
        }
    } while (!done);
    for (int i = 0; i < n; i++) {
        processes[i].turnaround_time = processes[i].burst_time + processes[i].waiting_time;
    }
}
```

```
void print_results(struct Process processes[], int n) {
    float total_waiting_time = 0;
    float total_turnaround_time = 0;
    printf("\nProcess ID   Burst Time  Waiting Time   Turnaround Time\n");
    for (int i = 0; i < n; i++) {
        total_waiting_time += processes[i].waiting_time;
        total_turnaround_time += processes[i].turnaround_time;
printf("Process[%d]    %d          %d           %d\n",
               processes[i].id,
               processes[i].burst_time,
               processes[i].turnaround_time,
               processes[i].waiting_time);
    }
    printf("\nAverage Waiting Time: %.6f\n", total_waiting_time / n);
    printf("Average Turnaround Time: %.6f\n", total_turnaround_time / n);
}
int main() {
    int n, quantum;
    printf("Enter Total Number of Processes: ");
    scanf("%d", &n);
    struct Process processes[n];
    for (int i = 0; i < n; i++) {
        processes[i].id = i + 1;
        printf("\nEnter Details of Process[%d]\n", processes[i].id);
        printf("Arrival Time: ");
        scanf("%d", &processes[i].arrival_time);
        printf("Burst Time: ");
        scanf("%d", &processes[i].burst_time);
        processes[i].remaining_time = processes[i].burst_time;
        processes[i].waiting_time = 0;
        processes[i].turnaround_time = 0;
```

```
    }
    printf("\nEnter Time Quantum: ");
    scanf("%d", &quantum);
    calculate_times(processes, n, quantum);
    print_results(processes, n);
    return 0;
}
```

Output :

```
Enter Total Number of Processes: 4

Enter Details of Process[1]
Arrival Time: 0
Burst Time: 4

Enter Details of Process[2]
Arrival Time: 1
Burst Time: 7

Enter Details of Process[3]
Arrival Time: 2
Burst Time: 5

Enter Details of Process[4]
Arrival Time: 3
Burst Time: 6

Enter Time Quantum: 3

Process ID   Burst Time  Waiting Time   Turnaround Time
Process[1]    4           13              9
Process[2]    7           21              14
Process[3]    5           16              11
Process[4]    6           18              12

Average Waiting Time: 11.500000
Average Turnaround Time: 17.000000
[cse32@localhost ~]$
```